University of Wah

Computer Science Department

Semester Spring - 2026

Lab Sessional

Block Chain

Course Code: CYS – 463L

Submitted To :

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NOTE:

-Attempt all the questions.

-All tasks will be submitted in a pdf file with file name [your name, registration No.].

-Maintain the order of the questions same as question paper in your answer sheet.

-Post Complete Report with Code, output on your LinkedIn & GitHub Profile.

-Understanding is part of paper.

Section 1 [LLO1, GA2, P1] [15 Marks]

Q1. Develop a Mini Student Web Portal consisting of multiple linked webpages. You are required to design and implement a multi-page website where all pages are connected through navigation.

PART 1 – Registration Page

File: registration.html

Requirements:

□ Create a student registration form with:

o Name

o Email

o Password

o Confirm Password

o Age

o Gender

Functional Requirements:

□ Apply Bootstrap styling

□ Perform JavaScript validation:

o Presence check (no empty fields)

o Email format validation

o Password match validation

o Age must be numeric (type check)

Navigation:

□ After successful validation → redirect to Dashboard Page

CODE:

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Registration</title>
    <meta charset="UTF-8">

    <!-- Bootstrap -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

    <!-- Custom CSS -->
    <link rel="stylesheet" href="style.css">
</head>

<body>

<div class="container mt-5">
    <h2 class="text-center">Student Registration</h2>

    <form id="regForm">

        <div class="mb-3">
```

```
    <label>Name</label>
    <input type="text" id="name" class="form-control">
</div>

<div class="mb-3">
    <label>Email</label>
    <input type="text" id="email" class="form-control">
</div>

<div class="mb-3">
    <label>Password</label>
    <input type="password" id="password" class="form-control">
</div>

<div class="mb-3">
    <label>Confirm Password</label>
    <input type="password" id="confirmPassword" class="form-control">
</div>

<div class="mb-3">
    <label>Age</label>
    <input type="text" id="age" class="form-control">
</div>

<div class="mb-3">
    <label>Gender</label><br>
    <input type="radio" name="gender" value="Male"> Male
```

```
            <input type="radio" name="gender" value="Female"> Female
        </div>

        <button type="submit" class="btn btn-primary w-100">Register</button>

    </form>
</div>

<script src="script.js"></script>

</body>
</html>
```

JAVA:

```
document.getElementById("regForm").addEventListener("submit", function(e) {
    e.preventDefault();

    let name = document.getElementById("name").value;
    let email = document.getElementById("email").value;
    let password = document.getElementById("password").value;
    let confirmPassword = document.getElementById("confirmPassword").value;
    let age = document.getElementById("age").value;

    // Presence check
    if (name === "" || email === "" || password === "" || confirmPassword === "" || age === "") {
        alert("All fields are required!");
        return;
    }
```

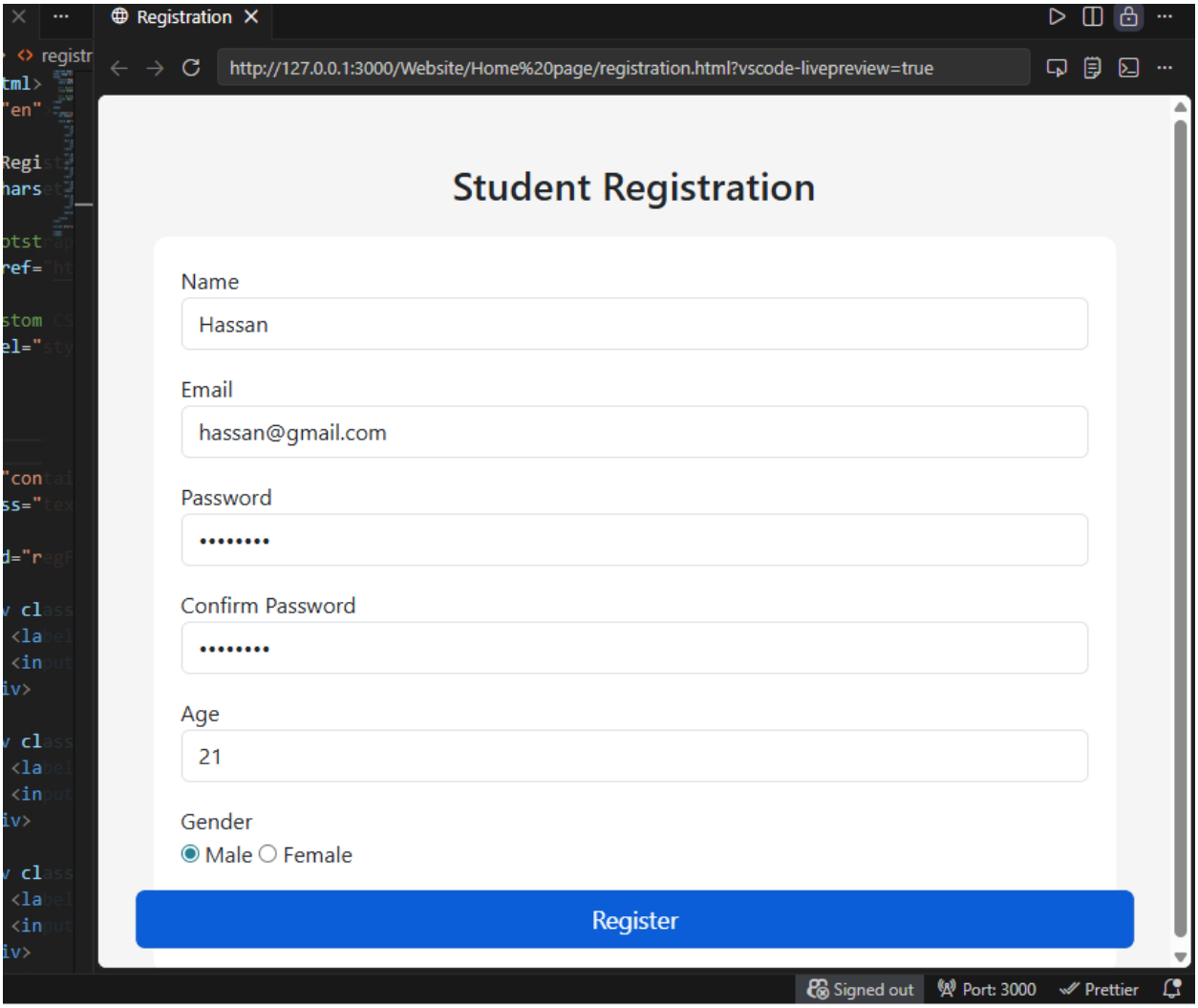
```
    // Email validation
    let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
    if (!email.match(emailPattern)) {
        alert("Invalid email format!");
        return;
    }

    // Password match
    if (password !== confirmPassword) {
        alert("Passwords do not match!");
        return;
    }

    // Age numeric check
    if (isNaN(age)) {
        alert("Age must be a number!");
        return;
    }

    // Success → Redirect
    alert("Registration Successful!");
    window.location.href = "dashboard.html";
});
```

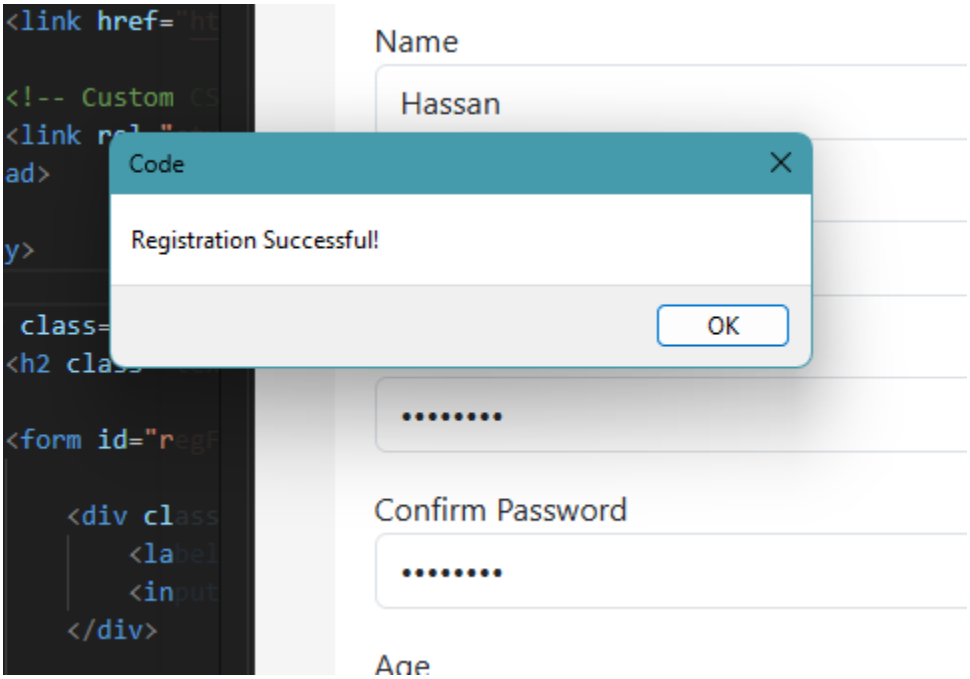
CSS:

```
body {
    background-color: #f5f5f5;
```

```
}

h2 {
    margin-bottom: 20px;
}

form {
    background: white;
    padding: 20px;
    border-radius: 10px;
}
```

OUTPUTS:


Registration
http://127.0.0.1:3000/Website/Home%20page/registration.html?vscode-livepreview=true
Student Registration
Name
Hassan
Email
hassan@gmail.com
Password
Confirm Password
Age
21
Gender
Male
Female
Register
Signed out
Port: 3000
Prettier


Name
Hassan
Code
Registration Successful!
OK
Confirm Password
Age

PART 2 – Dashboard Page

File: dashboard.html

Requirements:

□ Display:

o Welcome message (using JavaScript)

o Student Name (pass or simulate data)

□ Use Bootstrap Navbar & Cards

Functional Requirements:

□ Button: Show Current Date & Time

□ Navigation links to:

o Courses Page

o Feedback Page

o Home Page

CODE:
HTML:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <title>Dashboard</title>

    <meta charset="UTF-8">


    <!-- Bootstrap -->

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">


    <link rel="stylesheet" href="style.css">

</head>
```

```
<body>

<!-- Navbar -->
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
    <div class="container">
        <a class="navbar-brand" href="#">Student Portal</a>

        <div>
            <ul class="navbar-nav">
                <li class="nav-item"><a class="nav-link" href="index.html">Home</a></li>
                <li class="nav-item"><a class="nav-link" href="courses.html">Courses</a></li>
                <li class="nav-item"><a class="nav-link" href="feedback.html">Feedback</a></li>
            </ul>
        </div>
    </div>
</nav>

<div class="container mt-4">

    <!-- Welcome Message -->
    <h3 id="welcomeMsg"></h3>

<div class="row mt-4">

    <!-- Profile -->
    <div class="col-md-4">
        <a href="index.html" class="text-decoration-none text-dark">
```

```
        <div class="card p-3">
            <h5>Profile</h5>
            <p>View your details</p>
        </div>
    </a>
</div>

<!-- Courses -->
<div class="col-md-4">
    <a href="courses.html" class="text-decoration-none text-dark">
        <div class="card p-3">
            <h5>Courses</h5>
            <p>Manage your courses</p>
        </div>
    </a>
</div>

<!-- Feedback -->
<div class="col-md-4">
    <a href="feedback.html" class="text-decoration-none text-dark">
        <div class="card p-3">
            <h5>Feedback</h5>
            <p>Give your feedback</p>
        </div>
    </a>
</div>
```

```
</div>

    <!-- Date & Time Button -->
    <div class="mt-4">
        <button class="btn btn-success" onclick="showDateTime()">Show Date & Time</button>
        <p id="dateTime" class="mt-2"></p>
    </div>

</div>

<script src="dashboard.js"></script>

</body>
</html>
```

CSS:

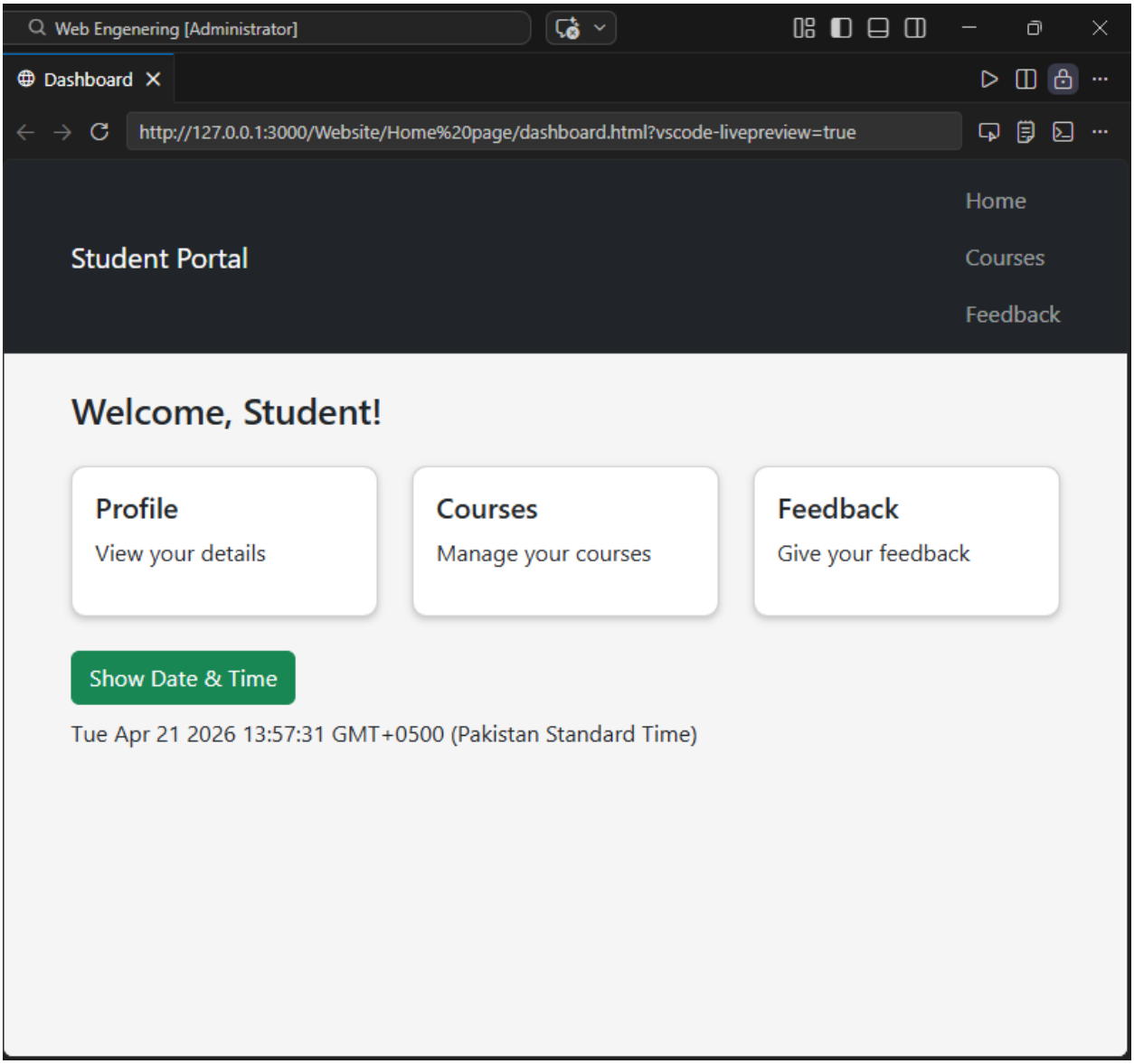
```
.card {
    border-radius: 10px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.2);
    transition: 0.3s;
}

.card:hover {
    transform: scale(1.05);
}
```

Java:

```
// Welcome message (simulate student name)
let studentName = "Student";
```

```
document.getElementById("welcomeMsg").innerText = "Welcome, " + studentName + "!";

// Show Date & Time
function showDateTime() {
    let now = new Date();
    document.getElementById("dateTime").innerText = now;
}
```

OUTPUTS:



PART 3 – Course Selection Page

File: courses.html

Requirements:

□ Display course list using checkboxes:

o Web Development

o Artificial Intelligence

o Database Systems

o Mobile App Development

Functional Requirements:

□ Button: Submit Courses

□ On click:

o Show selected courses using alert

Navigation:

□ Link back to Dashboard

CODE :
HTML :

```
<!DOCTYPE html>

<html lang="en">

<head>

    <title>Courses</title>

    <meta charset="UTF-8">


    <!-- Bootstrap -->

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
```

```
    <link rel="stylesheet" href="style.css">
</head>

<body>

<div class="container mt-5">

    <h2 class="text-center">Select Courses</h2>

    <form id="courseForm">

        <div class="form-check">
            <input class="form-check-input" type="checkbox" value="Web Development" id="c1">
            <label class="form-check-label">Web Development</label>
        </div>

        <div class="form-check">
            <input class="form-check-input" type="checkbox" value="Artificial Intelligence"
id="c2">
            <label class="form-check-label">Artificial Intelligence</label>
        </div>

        <div class="form-check">
            <input class="form-check-input" type="checkbox" value="Database Systems" id="c3">
            <label class="form-check-label">Database Systems</label>
        </div>
```

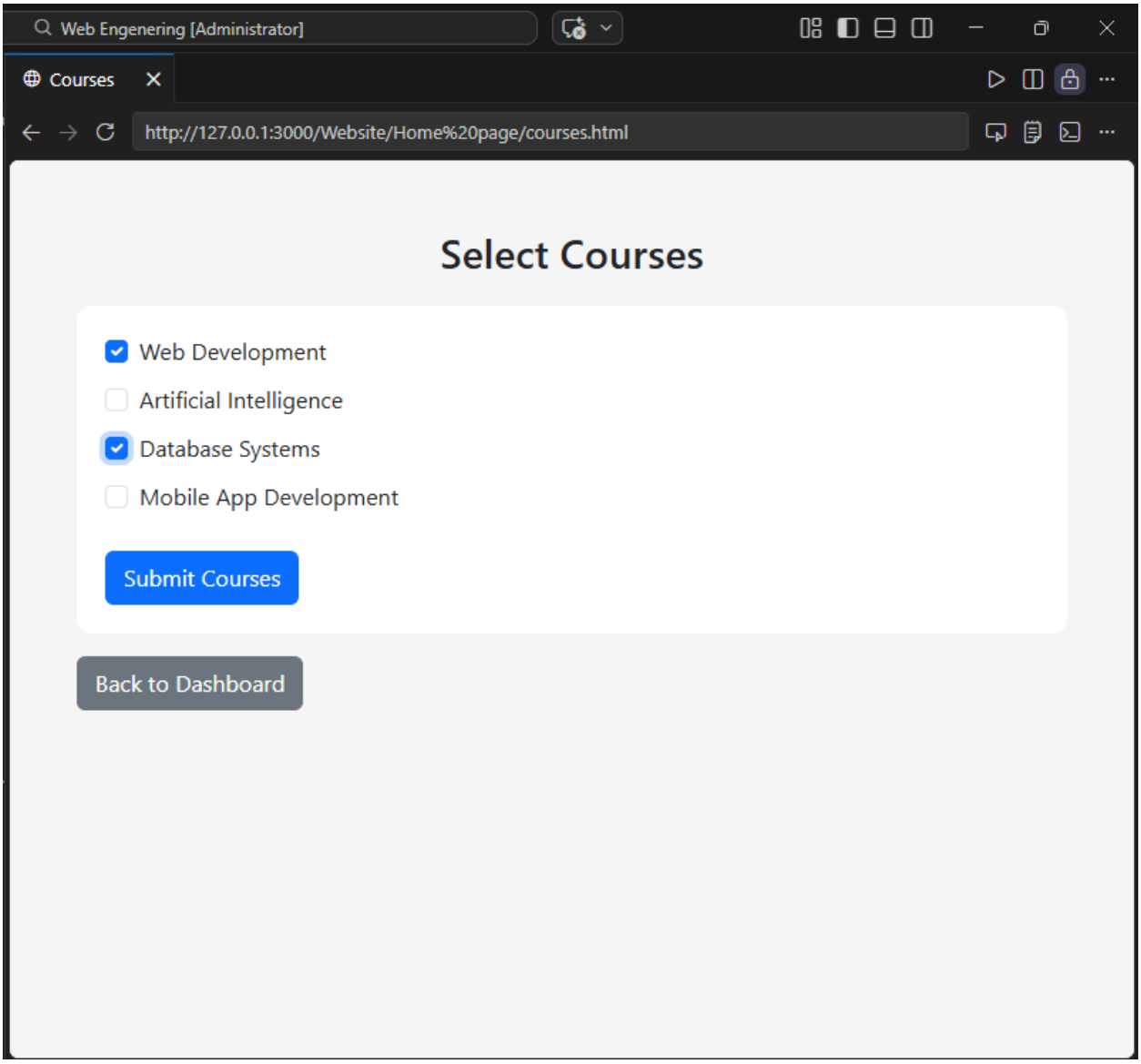
```
        <div class="form-check">
            <input class="form-check-input" type="checkbox" value="Mobile App Development"
id="c4">
            <label class="form-check-label">Mobile App Development</label>
        </div>

        <button type="button" class="btn btn-primary mt-3" onclick="submitCourses()">
            Submit Courses
        </button>

    </form>

    <!-- Back Button -->
    <div class="mt-3">
        <a href="dashboard.html" class="btn btn-secondary">Back to Dashboard</a>
    </div>

</div>

<script src="course.js"></script>

</body>
</html>
```

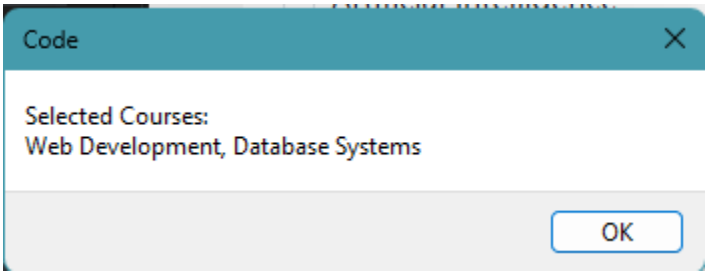
CSS:

```
.form-check {
    margin-bottom: 10px;
}
```

Java:

```
function submitCourses() {

    let selected = [];

    let courses = document.querySelectorAll("input[type='checkbox']:checked");

    courses.forEach(function(course) {
        selected.push(course.value);
    });

    if (selected.length === 0) {
        alert("Please select at least one course!");
        return;
    }
    alert("Selected Courses:\n" + selected.join(", "));
}
```





PART 4 – Feedback Page

File: feedback.html

Requirements:

□ Form fields:

o Name

o Comments

o Rating (Dropdown)

Functional Requirements:

 Apply validation (presence check)

 Show success message after submission

Navigation:

 Link back to Dashboard

CODE:

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Feedback</title>
    <meta charset="UTF-8">

    <!-- Bootstrap -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

    <link rel="stylesheet" href="style.css">
</head>

<body>

<div class="container mt-5">
```

```
<h2 class="text-center">Feedback Form</h2>

<form id="feedbackForm">

    <div class="mb-3">
        <label>Name</label>
        <input type="text" id="fname" class="form-control">
    </div>

    <div class="mb-3">
        <label>Comments</label>
        <textarea id="comments" class="form-control"></textarea>
    </div>

    <div class="mb-3">
        <label>Rating</label>
        <select id="rating" class="form-control">
            <option value="">Select Rating</option>
            <option>Excellent</option>
            <option>Good</option>
            <option>Average</option>
            <option>Poor</option>
        </select>
    </div>

    <button type="submit" class="btn btn-primary w-100">Submit Feedback</button>
```

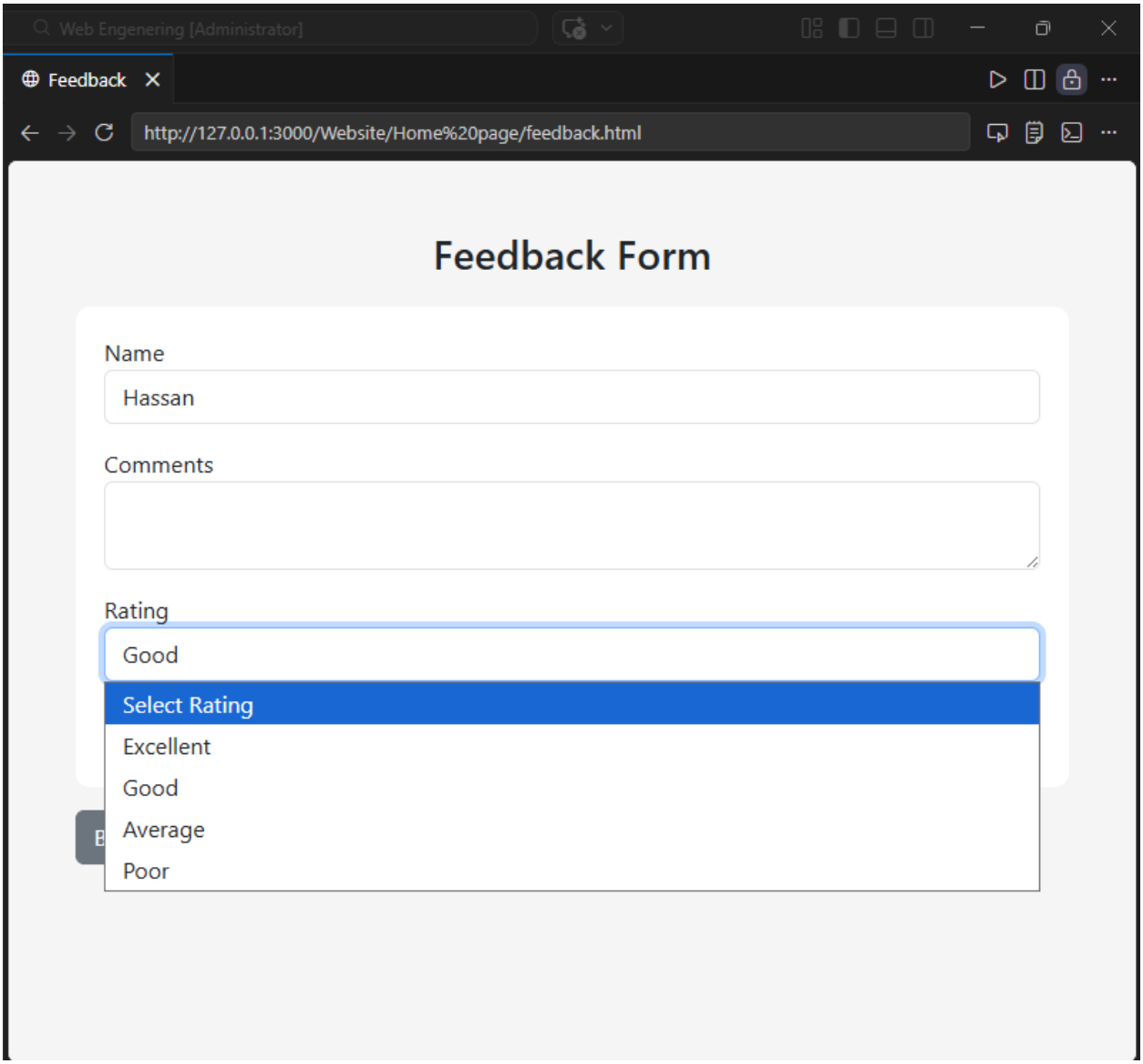
```
    </form>

    <!-- Success Message -->
    <p id="successMsg" class="text-success mt-3"></p>

    <!-- Back Button -->
    <div class="mt-3">
        <a href="dashboard.html" class="btn btn-secondary">Back to Dashboard</a>
    </div>

</div>

<script src="feedback.js"></script>

</body>
</html>
```

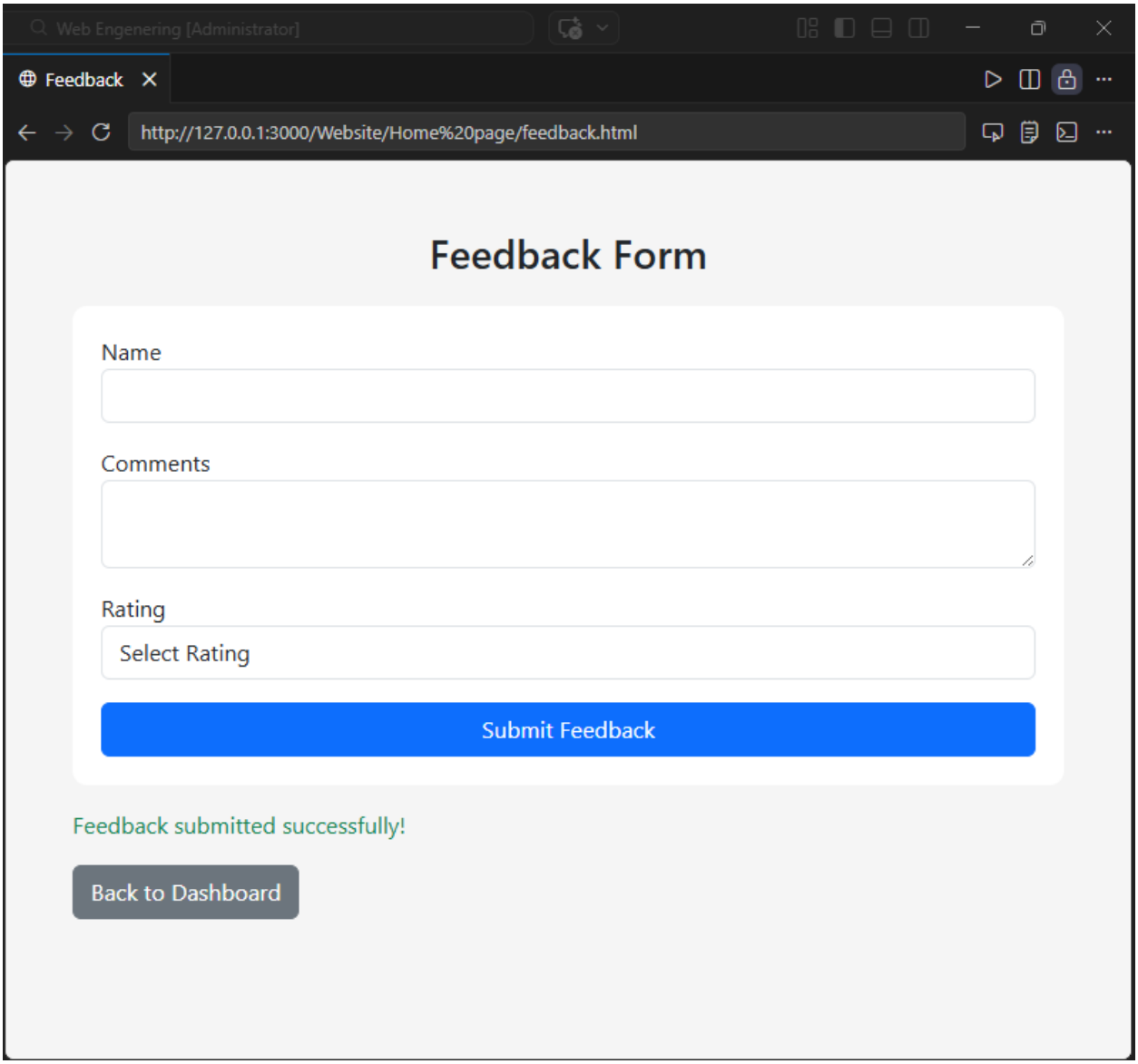
Java:

```
document.getElementById("feedbackForm").addEventListener("submit", function(e) {
    e.preventDefault();

    let name = document.getElementById("fname").value;
    let comments = document.getElementById("comments").value;
    let rating = document.getElementById("rating").value;

    // Presence check
    if (name === "" || comments === "" || rating === "") {
        alert("All fields are required!");
```

```
        return;
    }

    // Show success message
    document.getElementById("successMsg").innerText = "Feedback submitted successfully!";

    // Clear form
    document.getElementById("feedbackForm").reset();
});
```

OUTPUTS


Web Engenering [Administrator]
Feedback
http://127.0.0.1:3000/Website/Home%20page/feedback.html
Feedback Form
Name
Hassan
Comments
Rating
Good
Select Rating
Excellent
Good
Average
Poor



PART 5 – MAIN PAGE (Home Page)

File: index.html

Requirements:

□ Create a homepage with:

o Website title: Student Portal

o Navigation bar (Bootstrap)

Navigation Links:

□ Registration

 Dashboard

 Courses

 Feedback

Design:

 Apply CSS styling

 Add hover effects or basic UI enhancements

CODE:
HTML:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <title>Student Portal</title>

    <meta charset="UTF-8">


    <!-- Bootstrap -->

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">


    <link rel="stylesheet" href="style.css">

</head>


<body>


<!-- Navbar -->

<nav class="navbar navbar-expand-lg navbar-dark bg-primary">

    <div class="container">

        <a class="navbar-brand" href="#">Student Portal</a>
```

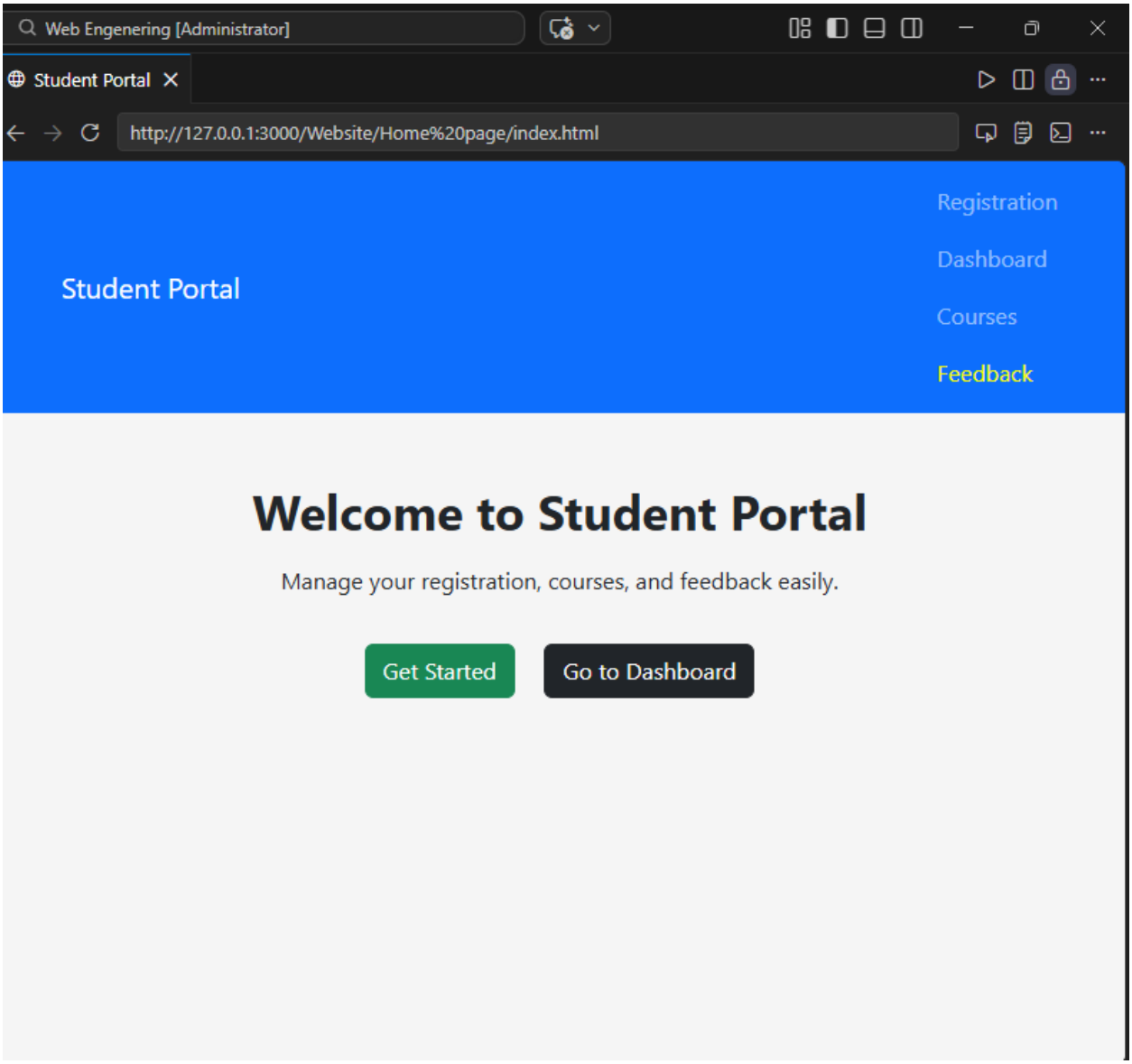
```
        <div>
            <ul class="navbar-nav">
                <li class="nav-item"><a class="nav-link"
href="registration.html">Registration</a></li>
                <li class="nav-item"><a class="nav-link" href="dashboard.html">Dashboard</a></li>
                <li class="nav-item"><a class="nav-link" href="courses.html">Courses</a></li>
                <li class="nav-item"><a class="nav-link" href="feedback.html">Feedback</a></li>
            </ul>
        </div>
    </div>
</nav>

<!-- Main Section -->
<div class="container text-center mt-5">

    <h1>Welcome to Student Portal</h1>
    <p class="mt-3">Manage your registration, courses, and feedback easily.</p>

    <div class="mt-4">
        <a href="registration.html" class="btn btn-success m-2">Get Started</a>
        <a href="dashboard.html" class="btn btn-dark m-2">Go to Dashboard</a>
    </div>

</div>

</body>
```

```
</html>
```

CSS:

```
.form-check {
    margin-bottom: 10px;
}
.navbar-nav .nav-link:hover {
    color: yellow !important;
}

h1 {
    font-weight: bold;
}

.btn {
    transition: 0.3s;
}

.btn:hover {
    transform: scale(1.1);
}
```

OUTPUTS:


Web Engenering [Administrator]
Student Portal
http://127.0.0.1:3000/Website/Home%20page/index.html
Student Portal
Registration
Dashboard
Courses
Feedback
Welcome to Student Portal
Manage your registration, courses, and feedback easily.
Get Started
Go to Dashboard